

Creation Date 09-May-2012

Revision Date 24-Jan-2024

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                               |
|---------------------------|-------------------------------|
| Product Description:      | <b>Styrene</b>                |
| Cat No. :                 | <b>A18481</b>                 |
| Synonyms                  | Ethenylbenzene                |
| Index No                  | 601-026-00-0                  |
| CAS No                    | 100-42-5                      |
| EC No                     | 202-851-5                     |
| Molecular Formula         | C <sub>8</sub> H <sub>8</sub> |
| REACH registration number | -                             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

[begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

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## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Flammable liquids

Category 3 (H226)

##### Health hazards

Aspiration Toxicity

Category 1 (H304)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Reproductive Toxicity

Category 2 (H361d)

Specific target organ toxicity - (single exposure)

Category 3 (H335)

Specific target organ toxicity - (repeated exposure)

Category 1 (H372)

##### Environmental hazards

Chronic aquatic toxicity

Category 3 (H412)

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

**Danger**

#### **Hazard Statements**

H226 - Flammable liquid and vapor

H304 - May be fatal if swallowed and enters airways

H332 - Harmful if inhaled

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

H361d - Suspected of damaging the unborn child

H372 - Causes damage to organs through prolonged or repeated exposure if inhaled

H412 - Harmful to aquatic life with long lasting effects

#### **Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician

P331 - Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

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## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                                    |
|-----------|----------|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   | 100-42-5 | EEC No. 202-851-5 | >95      | Flam. Liq. 3 (H226)<br>Acute Tox. 4 (H332)<br>Eye Irrit. 2 (H319)<br>Skin Irrit. 2 (H315)<br>STOT RE (H372)<br>Repr. 2 (H361d)<br>STOT SE 3 (H335)<br>Asp. Tox. 1 (H304)<br>Aquatic Chronic 3 (H412) |

REACH registration number

-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                   |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward. |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                   |
| Self-Protection of the First Aider | Use personal protective equipment as required.                                                                                                                                                      |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

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## 5.1. Extinguishing media

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

Water may be ineffective.

## 5.2. Special hazards arising from the substance or mixture

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

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Switzerland - Storage of hazardous substances

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union | The United Kingdom                                                                                                  | France                                                                                                                                                                                                                                                                                                                                      | Belgium                                                                                                                              | Spain                                                                                                                                                                       |
|-----------|----------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   |                | STEL: 250 ppm 15 min<br>STEL: 1080 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 430 mg/m <sup>3</sup> 8 hr | TWA / VME: 23.3 ppm (8 heures). indicative limit<br>TWA / VME: 100 mg/m <sup>3</sup> (8 heures). indicative limit<br>TWA / VME: 1000 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 46.6 ppm. restrictive limit<br>STEL / VLCT: 200 mg/m <sup>3</sup> . restrictive limit<br>STEL / VLCT: 1500 mg/m <sup>3</sup> .<br>Peau | TWA: 25 ppm 8 uren<br>TWA: 108 mg/m <sup>3</sup> 8 uren<br>STEL: 50 ppm 15 minuten<br>STEL: 216 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 40 ppm (15 minutos).<br>STEL / VLA-EC: 172 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 20 ppm (8 horas)<br>TWA / VLA-ED: 86 mg/m <sup>3</sup> (8 horas) |

| Component | Italy | Germany                                                                                                                                                                                                                                                   | Portugal                                       | The Netherlands | Finland                                                                                                                                      |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   |       | TWA: 20 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 86 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 20 ppm (8 Stunden). MAK<br>TWA: 86 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 40 ppm<br>Höhepunkt: 172 mg/m <sup>3</sup> | STEL: 40 ppm 15 minutos<br>TWA: 20 ppm 8 horas |                 | TWA: 20 ppm 8 tunteina<br>TWA: 86 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina<br>STEL: 430 mg/m <sup>3</sup> 15 minuutteina |

| Component | Austria                                                                                                                                           | Denmark                                                  | Switzerland                                                                                                                       | Poland                                                                           | Norway                                                                                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   | MAK-KZGW: 80 ppm 15 Minuten<br>MAK-KZGW: 340 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 20 ppm 8 Stunden<br>MAK-TMW: 85 mg/m <sup>3</sup> 8 Stunden | Ceiling: 25 ppm<br>Ceiling: 105 mg/m <sup>3</sup><br>Hud | STEL: 40 ppm 15 Minuten<br>STEL: 170 mg/m <sup>3</sup> 15 Minuten<br>TWA: 20 ppm 8 Stunden<br>TWA: 85 mg/m <sup>3</sup> 8 Stunden | STEL: 100 mg/m <sup>3</sup> 15 minutach<br>TWA: 50 mg/m <sup>3</sup> 8 godzinach | TWA: 25 ppm 8 timer<br>TWA: 105 mg/m <sup>3</sup> 8 timer<br>STEL: 37.5 ppm 15 minutter. value calculated<br>STEL: 131.25 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component | Bulgaria                                                      | Croatia                                                                                                                              | Ireland                                                                                                                                                                        | Cyprus | Czech Republic                                                                                                 |
|-----------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------|
| Styrene   | TWA: 85.0 mg/m <sup>3</sup><br>STEL : 215.0 mg/m <sup>3</sup> | kože<br>TWA-GVI: 100 ppm 8 satima.<br>TWA-GVI: 430 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 250 ppm 15 minutama.<br>STEL-KGVI: 1080 | TWA: 85 mg/m <sup>3</sup> 8 hr.<br>100% pure crystalline enzyme 100% pure crystalline enzyme<br>TWA: 20 ppm 8 hr.<br>STEL: 40 ppm 15 min<br>STEL: 170 mg/m <sup>3</sup> 15 min |        | TWA: 100 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 400 mg/m <sup>3</sup> |

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|  |  |                                |  |  |  |
|--|--|--------------------------------|--|--|--|
|  |  | mg/m <sup>3</sup> 15 minutama. |  |  |  |
|--|--|--------------------------------|--|--|--|

| Component | Estonia                                                                                                                                             | Gibraltar | Greece                                                                                      | Hungary                                                                                | Iceland                                                      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Styrene   | Nahk<br>TWA: 20 ppm 8 tundides.<br>TWA: 90 mg/m <sup>3</sup> 8 tundides.<br>STEL: 50 ppm 15 minutites.<br>STEL: 200 mg/m <sup>3</sup> 15 minutites. |           | STEL: 250 ppm<br>STEL: 1050 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 425 mg/m <sup>3</sup> | STEL: 172 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 86 mg/m <sup>3</sup> 8 órában. AK | STEL: 25 ppm<br>STEL: 105 mg/m <sup>3</sup><br>Skin notation |

| Component | Latvia                                                  | Lithuania                                                                                                                                                                             | Luxembourg | Malta | Romania                                                                                                                 |
|-----------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|-------------------------------------------------------------------------------------------------------------------------|
| Styrene   | STEL: 30 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> | TWA: 20 ppm IPRD<br>TWA: 90 mg/m <sup>3</sup> IPRD<br>TWA: 10 ppm IPRD for planning of new facilities or replacing the old ones<br>Oda<br>STEL: 50 ppm<br>STEL: 200 mg/m <sup>3</sup> |            |       | TWA: 12 ppm 8 ore<br>TWA: 50 mg/m <sup>3</sup> 8 ore<br>STEL: 35 ppm 15 minute<br>STEL: 150 mg/m <sup>3</sup> 15 minute |

| Component | Russia                                                      | Slovak Republic                                                            | Slovenia                                                                                                                    | Sweden                                                                                                                                                                | Turkey |
|-----------|-------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Styrene   | TWA: 10 mg/m <sup>3</sup> 2399<br>MAC: 30 mg/m <sup>3</sup> | Ceiling: 200 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 86 mg/m <sup>3</sup> | TWA: 20 ppm 8 urah<br>TWA: 86 mg/m <sup>3</sup> 8 urah<br>STEL: 40 ppm 15 minutah<br>STEL: 172 mg/m <sup>3</sup> 15 minutah | Indicative STEL: 20 ppm 15 minuter<br>Indicative STEL: 86 mg/m <sup>3</sup> 15 minuter<br>TLV: 10 ppm 8 timmar. NGV<br>TLV: 43 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud |        |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Spain                                                                                                                    | Germany                                                                                                                                                                                                                           |
|-----------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   |                |                | Styrene: 0.02 mg/L venous blood Before the beginning of the next shift<br>Styrene: 0.04 mg/L urine end of shift<br>Mandelic acid and Phenylglyoxyl: 400 mg/g creatinine urine end of shift, preferably at end of workweek<br>Mandelic acid: 300 mg/g creatinine urine Before the beginning of the next shift<br>Styrene: 0.55 mg/L venous blood end of shift<br>Mandelic acid: 800 mg/g creatinine urine end of shift<br>Phenylglyoxylic acid: 240 mg/g creatinine urine end of shift<br>Phenylglyoxylic acid: 100 mg/g creatinine urine prior to shift | Mandelic acid plus Phenylglyoxylic acid: 400 mg/g Creatinine end of shift<br>Styrene: 0.2 mg/L venous blood end of shift | Mandelic acid plus Phenylglyoxylic acid: 600 mg/g Creatinine urine (end of shift )<br>Mandelic acid plus Phenylglyoxylic acid: 600 mg/g Creatinine urine (for long-term exposures: at the end of the shift after several shifts ) |

| Component | Italy | Finland                                                     | Denmark | Bulgaria                                                 | Romania                                               |
|-----------|-------|-------------------------------------------------------------|---------|----------------------------------------------------------|-------------------------------------------------------|
| Styrene   |       | MAPGA: 1.2 mmol/L urine in the morning after a working day. |         | Mandelic acid and Phenylglyoxylic acid - total: 600 mg/g | Mandelic acid: 800 mg/g Creatinine urine end of shift |

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|  |  |                                                                |  |                                                                                                              |                                                                                                                                                                                                                                  |
|--|--|----------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | MAPGA equals sum of urinary Mandelic and Phenylglyoxylic acids |  | Creatinine urine at the end of exposure or end of work shift, in remote exposure - after several work shifts | Mandelic acid: 300 mg/g<br>Creatinine urine beginning of next shift<br>Phenylglyoxylic acid: 100 mg/g Creatinine urine end of shift<br>Styrene: 0.55 mg/L blood end of shift<br>Styrene: 0.02 mg/L blood beginning of next shift |
|--|--|----------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component | Gibraltar | Latvia                                                                                           | Slovak Republic                                                                                                                                                                                                 | Luxembourg | Turkey |
|-----------|-----------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Styrene   |           | Mandelic acid: 0.8 g/g<br>Creatinine urine end of shift<br>Styrene: 0.55 mg/L blood end of shift | Mandelic acid and Phenylglycolic acid: 600 mg/g creatinine urine after all work shifts for long-term exposure<br>Mandelic acid and Phenylglycolic acid: 600 mg/g creatinine urine end of exposure or work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

## Predicted No Effect Concentration (PNEC)

See values below.

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

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**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Maintain adequate ventilation Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                      |                                          |
|------------------------------------------------|--------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                               |                                          |
| <b>Appearance</b>                              | Colorless                            |                                          |
| <b>Odor</b>                                    | pungent                              |                                          |
| <b>Odor Threshold</b>                          | No data available                    |                                          |
| <b>Melting Point/Range</b>                     | -31 °C / -23.8 °F                    |                                          |
| <b>Softening Point</b>                         | No data available                    |                                          |
| <b>Boiling Point/Range</b>                     | 145 - 146 °C / 293 - 294.8 °F        | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Flammable                            | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                       | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.1<br><b>Upper</b> 6.1 |                                          |
| <b>Flash Point</b>                             | 31 °C / 87.8 °F                      | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 490 °C / 914 °F                      |                                          |
| <b>Decomposition Temperature</b>               | No data available                    |                                          |
| <b>pH</b>                                      | No information available             |                                          |
| <b>Viscosity</b>                               | 0.695 mPa.s at 25 °C                 |                                          |
| <b>Water Solubility</b>                        | 0.3 mg/L (20°C)                      |                                          |
| <b>Solubility in other solvents</b>            | Soluble: Alcohols, Diethyl ether     |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                      |                                          |
| <b>Component</b>                               | <b>log Pow</b>                       |                                          |
| Styrene                                        | 2.96                                 |                                          |
| <b>Vapor Pressure</b>                          | 7 mbar @ 20 °C                       |                                          |
| <b>Density / Specific Gravity</b>              | 0.906                                |                                          |
| <b>Bulk Density</b>                            | Not applicable                       | Liquid                                   |
| <b>Vapor Density</b>                           | 1.22                                 | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)              |                                          |

### 9.2. Other information

|                                         |                                        |
|-----------------------------------------|----------------------------------------|
| <b>Molecular Formula</b>                | C8 H8                                  |
| <b>Molecular Weight</b>                 | 104.15                                 |
| <b>Explosive Properties</b>             | explosive air/vapour mixtures possible |
| <b>Self-accelerating polymerisation</b> | 52°C (packages up to 50kg)             |



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temperature (SAPT)

Inhibitor level > 8 ppm

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

Hazardous polymerization may occur. Hazardous polymerization may occur upon depletion of inhibitor.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Excess heat. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. Temperatures above 40°C.

### 10.5. Incompatible materials

Acids. Halogenated compounds. Copper alloys. Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

| Component | LD50 Oral | LD50 Dermal               | LC50 Inhalation              |
|-----------|-----------|---------------------------|------------------------------|
| Styrene   | -         | LD50 > 2000 mg/kg ( Rat ) | LC50 = 11.7 mg/L ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Category 2

#### (c) serious eye damage/irritation;

Category 2

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

#### (f) carcinogenicity;

Based on available data, the classification criteria are not met

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Styrene   |    |    |         | Group 2A |

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|                                                      |                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------|
| (g) reproductive toxicity;<br>Reproductive Effects   | Category 2<br>Experiments have shown reproductive toxicity effects on laboratory animals. |
| (h) STOT-single exposure;<br>Results / Target organs | Category 3<br>Respiratory system.                                                         |
| (i) STOT-repeated exposure;<br>Target Organs         | Category 1<br>Ears, Central nervous system (CNS).                                         |
| (j) aspiration hazard;<br>Other Adverse Effects      | Category 1<br>The toxicological properties have not been fully investigated.              |
| Symptoms / effects, both acute and delayed           | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.      |

## 11.2. Information on other hazards

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Endocrine Disrupting Properties | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Do not empty into drains. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component | Freshwater Fish                                                                                                                                                                                                                                        | Water Flea                                | Freshwater Algae                                                                                                                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Styrene   | LC50: 19.03 - 33.53 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 58.75 - 95.32 mg/L, 96h static (Poecilia reticulata)<br>LC50: 6.75 - 14.5 mg/L, 96h static (Pimephales promelas)<br>LC50: 3.24 - 4.99 mg/L, 96h flow-through (Pimephales promelas) | EC50: 3.3 - 7.4 mg/L, 48h (Daphnia magna) | EC50: 0.15 - 3.2 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 0.46 - 4.3 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.72 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: = 1.4 mg/L, 72h (Pseudokirchneriella subcapitata) |

| Component | Microtox                                         | M-Factor |
|-----------|--------------------------------------------------|----------|
| Styrene   | = 5.4 mg/L EC50 Photobacterium phosphoreum 5 min |          |

### 12.2. Persistence and degradability

#### Persistence

#### Degradation in sewage treatment plant

Insoluble in water, Persistence is unlikely, based on information available.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

May have some potential to bioaccumulate

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Styrene   | 2.96    | 13.5 dimensionless            |

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## 12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and floats on water The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces . Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility.

## 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

| Component | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-----------|------------------------------------------|--------------------------------------------------|
| Styrene   | Group I Chemical                         | High Exposure Concern                            |

## 12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

#### **Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN2055

#### 14.2. UN proper shipping name

STYRENE MONOMER, STABILIZED

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

III

### ADR

#### 14.1. UN number

UN2055

#### 14.2. UN proper shipping name

STYRENE MONOMER, STABILIZED

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

III

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## IATA

**14.1. UN number** UN2055  
**14.2. UN proper shipping name** STYRENE MONOMER, STABILIZED  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** III

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** Inhibitors have been added to stabilize this product. Inhibitor levels should be maintained. Hazardous polymerization may occur upon depletion of inhibitor.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Styrene   | 100-42-5 | 202-851-5 | -      | -   | X     | X    | KE-35342 | X    | X    |

| Component | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Styrene   | 100-42-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Styrene   | 100-42-5 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Styrene   | 100-42-5 | Not applicable                                                                            | Not applicable                                                                           |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**

Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

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Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------|---------------------------------------|-------------------------|
| Styrene   | WGK2                                  |                         |

| Component | France - INRS (Tables of occupational diseases)      |
|-----------|------------------------------------------------------|
| Styrene   | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                   | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Styrene<br>100-42-5 ( >95 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H304 - May be fatal if swallowed and enters airways  
H332 - Harmful if inhaled  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H361d - Suspected of damaging the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H412 - Harmful to aquatic life with long lasting effects  
H226 - Flammable liquid and vapor  
H302 - Harmful if swallowed  
H312 - Harmful in contact with skin  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

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**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
| <b>Creation Date</b>    | 09-May-2012                                        |
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| <b>Revision Summary</b> | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

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**End of Safety Data Sheet**